

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_csv("netflix_titles.csv")
```

```
In [3]: df.head()
```

Out[3]:

	show_id	type	title	director	cast	country	date_added	release_year	rating
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabane...	South Africa	September 24, 2021	2021	TV MA
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV MA
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV MA
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV MA

Dataset Structure

Dataset Information

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Dataset Shape

```
In [5]: df.shape
```

```
Out[5]: (8807, 12)
```

Column Names

```
In [6]: df.columns
```

```
Out[6]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
              'release_year', 'rating', 'duration', 'listed_in', 'description'],
              dtype='object')
```

Check Missing Values

```
In [7]: df.isnull().sum()
```

```
Out[7]: show_id      0
        type        0
        title       0
        director    2634
        cast        825
        country     831
        date_added  10
        release_year 0
        rating      4
        duration    3
        listed_in   0
        description 0
        dtype: int64
```

Summary Statistics

```
In [8]: df.describe()
```

Out[8]:

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

```
In [9]: df.describe(include='object')
```

Out[9]:

	show_id	type	title	director	cast	country	date_added	rating	du
count	8807	8807	8807	6173	7982	7976	8797	8803	
unique	8807	2	8807	4528	7692	748	1767	17	
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	TV-MA	1 s
freq	1	6131	1	19	19	2818	109	3207	

Data Cleaning

Remove duplicates

```
In [11]: df.drop_duplicates(inplace=True)
```

Fill Missing Values

```
In [12]: df['country'].fillna("Unknown", inplace=True)

df['director'].fillna("Not Available", inplace=True)
```

Correlation Analysis

Select Numerical Columns

```
In [14]: numeric_df = df.select_dtypes(include=['int64', 'float64'])
```

Correlation Matrix

```
In [15]: corr = numeric_df.corr()
corr
```

```
Out[15]:
```

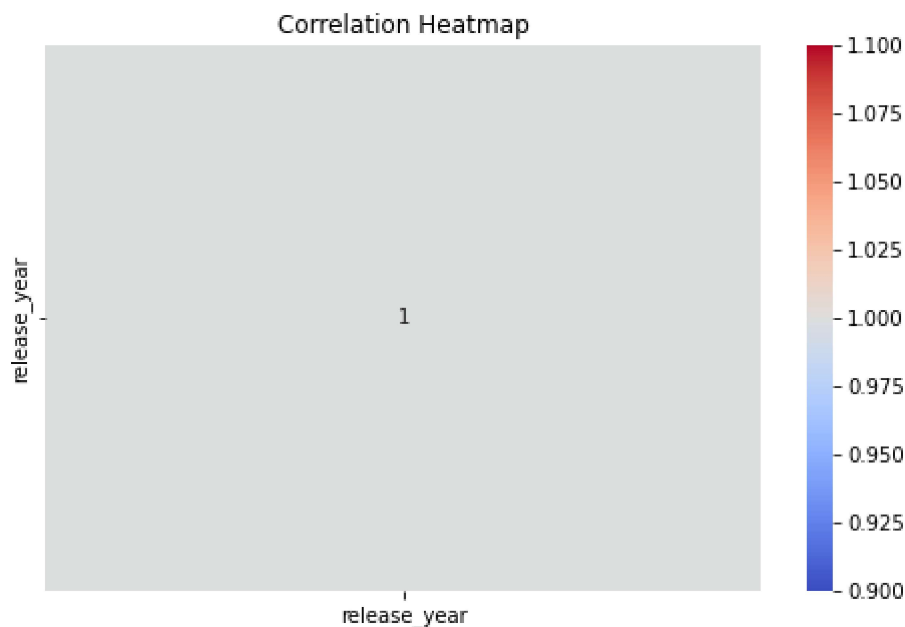
	release_year
release_year	1.0

Heatmap Visualization

```
In [17]: plt.figure(figsize=(8,5))

sns.heatmap(corr,
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```

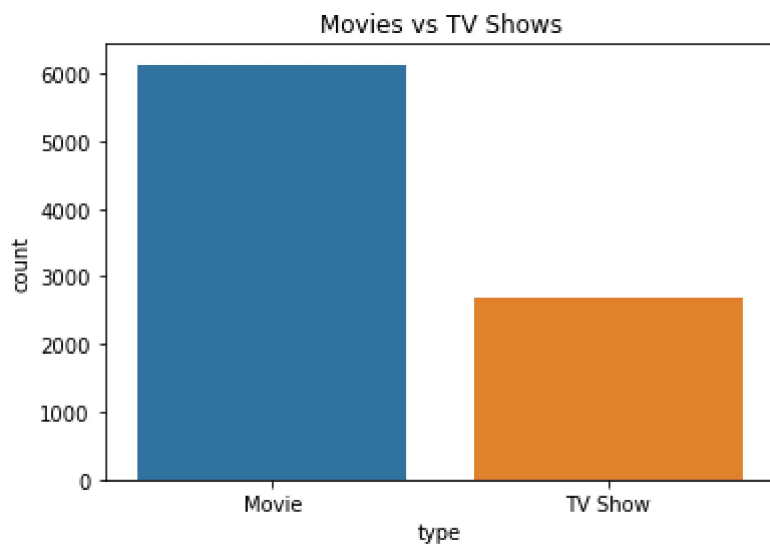


Trend Identification

Trend 1 — Movies vs TV Shows

```
In [19]: sns.countplot(x='type', data=df)

plt.title("Movies vs TV Shows")
plt.show()
```



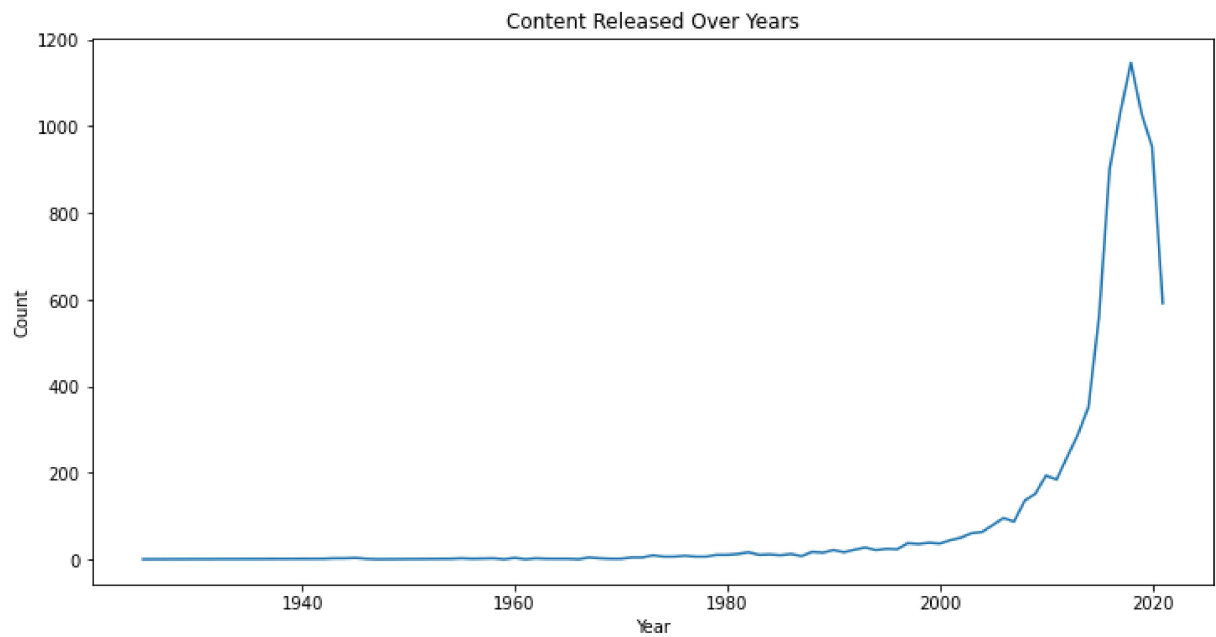
Trend 2 — Releases by Year

```
In [20]: plt.figure(figsize=(12,6))

df['release_year'].value_counts()\
.sort_index()\
.plot()
```

```
plt.title("Content Released Over Years")
plt.xlabel("Year")
plt.ylabel("Count")

plt.show()
```



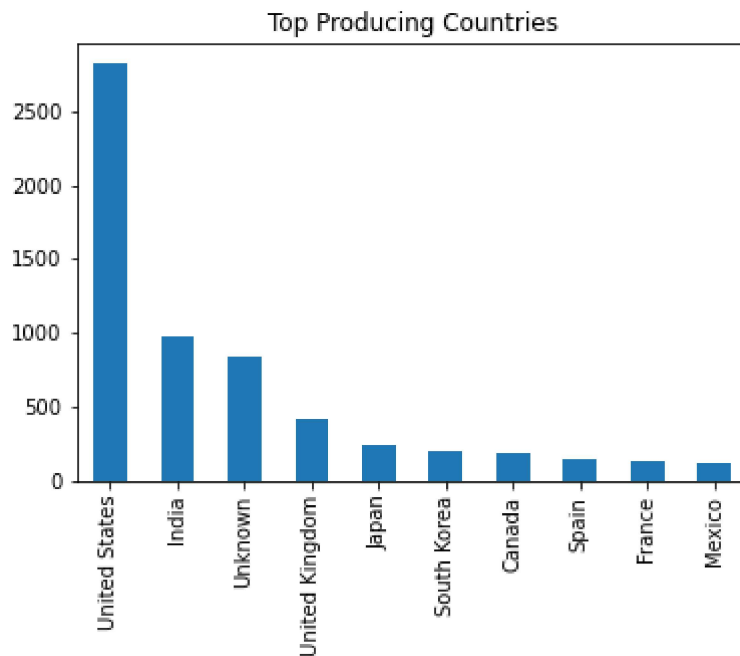
Trend 3 — Top 10 Countries

```
In [21]: top_country = df['country'].value_counts().head(10)

top_country.plot(kind='bar')

plt.title("Top Producing Countries")

plt.show()
```



Trend 4 — Content Ratings

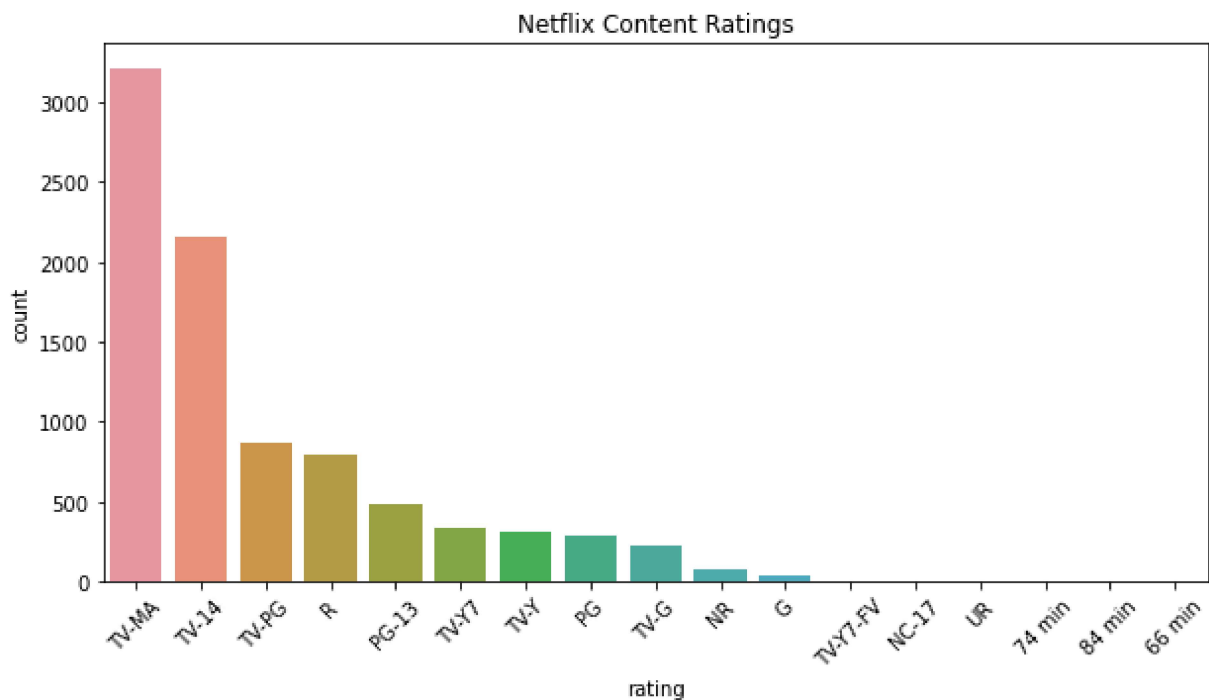
```
In [22]: plt.figure(figsize=(10,5))

sns.countplot(
    x='rating',
    data=df,
    order=df['rating'].value_counts().index
)

plt.xticks(rotation=45)

plt.title("Netflix Content Ratings")

plt.show()
```



Trend 5 — Monthly Additions

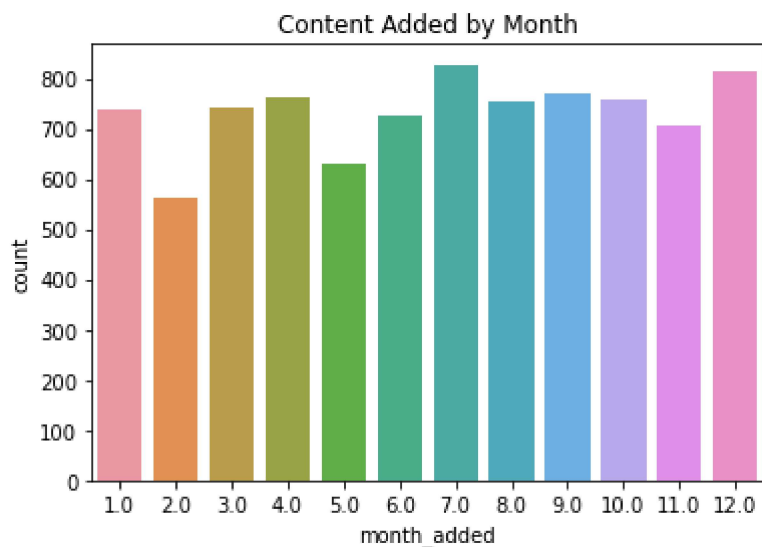
```
In [23]: df['date_added'] = pd.to_datetime(
df['date_added']
)

df['month_added'] = df['date_added'].dt.month
```

```
In [24]: sns.countplot(
x='month_added',
data=df
)

plt.title("Content Added by Month")

plt.show()
```



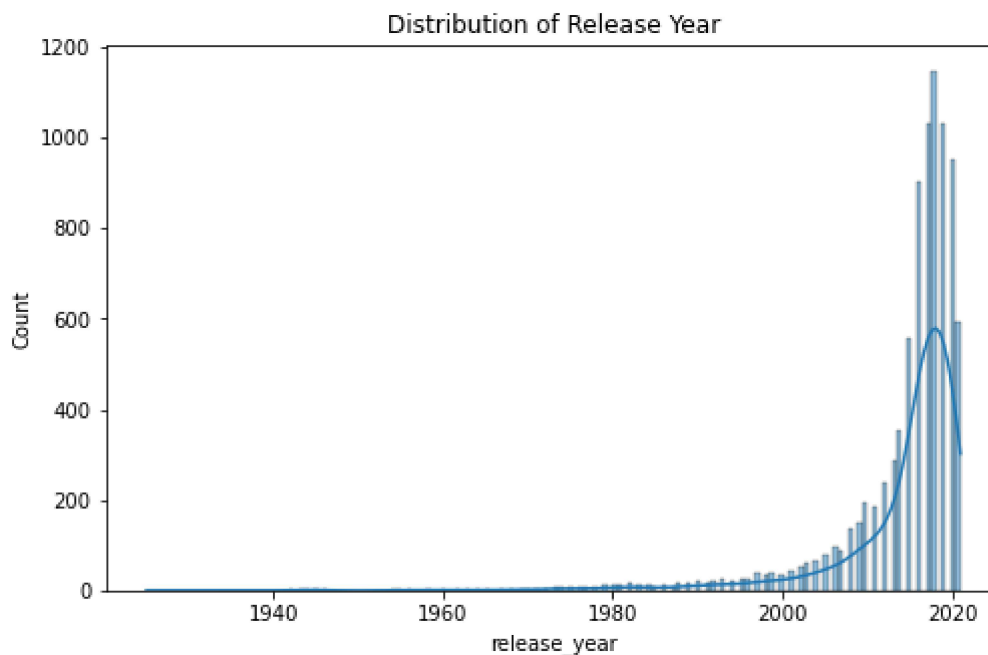
Histogram

```
In [25]: plt.figure(figsize=(8,5))

sns.histplot(df['release_year'],
kde=True)

plt.title("Distribution of Release Year")

plt.show()
```

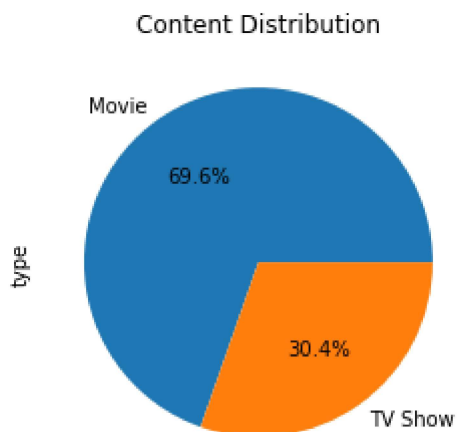


Pie Chart

```
In [26]: df['type'].value_counts().plot(
kind='pie',
autopct='%1.1f%%'
)

plt.title("Content Distribution")

plt.show()
```



Insights

Netflix hosts more Movies than TV Shows.

Content production increased sharply after 2015.

USA produces maximum Netflix content.

TV-MA rating dominates the platform.

CONCLUSION

The Netflix dataset EDA revealed major content trends. Movies dominate the platform. Content production rapidly increased after 2015. USA contributes most content. TV-MA is the most frequent rating. Correlation analysis helped understand relationships between numerical features.

In []: